Raghav Chandna

AI/ML and Full-Stack Developer working across spatio-temporal modelling, graph neural networks, sensor analytics, and research-oriented software development.

Rewari, India
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raghav-chandna
Portfolio

Present Position

AI Developer, South Africa Consortium for Air Quality Management (SACAQM),
University of the Witwatersrand, Johannesburg, South Africa 2024 – Present

Education

B.Sc. (Hons) Physics, University of Delhi, India July 2024
First Division; Thesis Grade: A+ (Excellent)
Thesis: **Optical Switches: An Application of Optomechanics**

- Formulated the Hamiltonian of a cavity–mirror optomechanical system using creation and annihilation operators.
- Applied Heisenberg equations of motion to derive a nonlinear cubic relation in photon number.
- Analysed optical bistability and decay-rate reduction using an improved mirror design.

Publications

2026 *Spatio-temporal Modeling Using a Cost-Effective System for Air Quality Monitoring.*
Manal Karmoude, **Raghav Chandna**, *et al.*
IEEE Conference on Artificial Intelligence (CAI 2026).
10.1109/CAI68641.2026.11536651.

2025 *A Scalable Reinforcement Learning Framework for Wireless Sensor Networks: Spatiotemporal Cluster-Based Multi-Agent Reinforcement Learning.*
Isaiah Chiraira, **Raghav Chandna**, Bruce Mellado, *et al.*
IEEE ACDSA 2025.
10.1109/ACDSA65407.2025.11166147.

2025 *Efficient Transfer Learning for Time-Series Forecasting Using Wi-Fi Data.*
Raghav Chandna, Bruce Mellado, *et al.*
IEEE IMITEC 2025.
10.1109/IMITEC67386.2025.11410454.

2025 *Transfer Learning from Wi-Fi Access Point Data to Air Quality Monitoring: A Spatial-Temporal Graph-Based Approach.*
Raghav Chandna, Bruce Mellado, *et al.*
IEEE ICECET 2025.
10.1109/ICECET63943.2025.11472283.

2025 *Network Load Forecasting via SARIMA–BiLSTM Transfer Learning.*
Raghav Chandna, Bruce Mellado, *et al.*
Springer DMBD 2025.
10.1007/978-92-0229-4_19.

Professional Experience

AI Developer, SACAQM, University of the Witwatersrand Sept 2024 – Present

- Develop spatio-temporal forecasting workflows for Wi-Fi access-point activity and air-quality sensor data using graph neural networks and recurrent neural networks.
- Build GraphSAGE-based pretraining and transfer-learning pipelines between Wi-Fi data and air-quality targets, including PM2.5 and PM4.0.
- Design and execute Weights & Biases hyperparameter experiments for graph-based and LSTM forecasting models.
- Contribute to React-based environmental monitoring dashboards featuring sensor readings, station maps, pollutant trends, threshold exceedances, and device-status indicators.
- Develop and maintain backend services and REST APIs using Node.js, Express, MongoDB, and Mongoose for sensor, station, device, and monitoring workflows.
- Contribute to Python-based forecasting services using time-series models, including SARIMA and Conv-LSTM approaches, with data validation and historical evaluation workflows.
- Implement AI-assisted conversational functionality for air-quality queries and insights using an LLM/agent API.
- Develop UI components for the official SACAQM Android application using Kotlin and support dashboard maintenance across frontend and backend components.
- Support scheduled prediction and sensor-alert workflows, technical documentation, and biweekly stakeholder reports and presentations.

Selected Technical Work

Environmental Monitoring and Compliance Dashboards

- Developed and maintained dashboard components for environmental telemetry, including pollutant visualisations, station maps, sensor status, and exceedance indicators.
- Worked with React, JavaScript, mapping libraries, Chart.js, and Material UI to present sensor data and monitoring information.
- Contributed to dashboard interfaces for different monitoring contexts, including environmental compliance and private sensor monitoring.

Spatio-Temporal Forecasting and Transfer Learning

- Applied graph-based and temporal models to Wi-Fi access-point and air-quality datasets.
- Built GraphSAGE-based pretraining workflows to investigate transfer learning from Wi-Fi data to air-quality targets, including PM2.5 and PM4.0.
- Experimented with GNN, LSTM, SARIMA, and hybrid forecasting approaches, using Python, PyTorch, and PyTorch Geometric.
- Used Weights & Biases to organise experiments and evaluate model performance across hyperparameter settings.

Air-Quality Forecasting Microservice

- Contributed to a Python-based forecasting service using FastAPI and time-series modelling approaches.

- Worked on prediction workflows, historical evaluation, data validation, and handling sparse or missing sensor data.
- Explored SARIMA and Conv-LSTM approaches for multi-sensor air-quality forecasting.

Sensor APIs, Data Management and Automated Alerts

- Developed and maintained backend components using Node.js, Express, MongoDB, and Mongoose for sensor, station, and device data.
- Worked on REST API endpoints, protected routes, and workflows for retrieving and managing monitoring data.
- Contributed scheduled device-status checks and alert workflows for offline sensors and threshold violations.

Agentic AI and Conversational Monitoring

- Integrated LLM/agent API functionality into air-quality monitoring workflows to support conversational queries and AI-generated insights.
- Worked on backend routes and event-tracking components supporting interactions between the assistant and monitoring platform.

Android Application Development

- Developed user-interface components for the official SACAQM Android application using Kotlin and Android Studio.
- Worked across application UI development and integration with monitoring-related functionality.

Deployment and Development Infrastructure

- Worked with Git-based development workflows, Docker, Linux, and environment configuration.
- Gained experience with deployment configuration and database connectivity troubleshooting in containerised environments.

Technical Skills

Languages	Python, Kotlin, Java, JavaScript, HTML/CSS, SQL, Bash
Frontend and Mobile	React, Vite, Mapbox GL, Leaflet, Chart.js, Material UI, Android Studio, Kotlin
Backend and APIs	Node.js, Express, FastAPI, REST APIs, Uvicorn, scheduled jobs and alert workflows
Machine Learning and AI	PyTorch, PyTorch Geometric, TensorFlow/Keras, Scikit-learn, Graph Neural Networks (GraphSAGE, GCN), LSTM, Conv-LSTM, SARIMA, transfer learning, time-series forecasting, reinforcement learning fundamentals, hyperparameter optimisation
Data Analysis	NumPy, Pandas, Matplotlib, Seaborn, exploratory data analysis, feature engineering, preprocessing, cross-validation, statistical analysis, model evaluation
Databases and Tools	MongoDB Atlas, Mongoose, Git/GitHub, Weights & Biases, Docker, Linux/Ubuntu, VS Code, Jupyter Notebook, Google Colab, Android Studio
Cloud and Deployment	Basic AWS EC2/S3, model inference pipelines, Linux system configuration, reproducible experiments
Other	Technical writing, research documentation, UI/UX design, stakeholder reporting, Agile collaboration